Concept Drift Detection in Polarized News Streams: An Online Machine Learning Approach

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Abstract

Online political discourse constantly evolves, creating potential concept drift that degrades static NLP models. This paper develops and evaluates two complementary approaches for drift detection in news streams sourced from the GDELT database spanning the 2024 U.S. Presidential Election.

In the first part (Classification-based drift), we compare various vectorization and online classifier strategies on synthetic and real-world data. We use the ADWIN drift detector to detect drift in the stream of errors produced by classifiers.

In the second part (Distributional-based drift), we apply an unsupervised distributional drift pipeline using daily and weekly article aggregates. Cosine distances are compared for consecutive windows in TF-IDF and embedding spaces, and the resulting distance streams are monitored with drift detectors. This avoids reliance on classification accuracy.

We show that given sufficient amount of data both approaches are able to detect shifts aligned with some major political events.

Introduction

News media streams reflect rapidly evolving political language: terms that signal one ideology may be adopted by opposing factions within months. This creates *concept drift*: a change in the joint distribution $P(X, y)$ over time that degrades static NLP models [1]. Batch-trained classifiers work on a fixed mapping from features to labels: when semantics shift around major political events, they cannot adapt without retraining.

This paper develops and evaluates two complementary approaches for drift detection in online news streams. Our contributions are:

1. A GDELT data scraping pipeline for politically polarized articles spanning the 2024 U.S. Presidential Election.
2. A synthetic benchmark generator simulating abrupt, gradual, and recurring drifts with known ground truth for controlled evaluation.
3. **Part I**: A systematic comparison of classification-based drift detection using TF-IDF [2] and dense LLM embeddings with online classifiers.
4. **Part II**: An unsupervised distributional drift detection pipeline that compares consecutive TF-IDF and embedding-based window representations using cosine distance, then applies drift detectors across three data stream types.
5. Empirical evidence that both approaches yield promising results and prove to be a useful tool when working with media streams.

The remainder of this paper is organized as follows. Section 2 details the data acquisition and preprocessing, synthetic generation, dynamic text vectorization techniques and methodologies for the supervised classification and unsupervised distributional drift approaches. Section 3 presents conducted experiments and their results across both synthetic and real-world streams. Finally, Section 4 discusses key findings, limitations, and future directions.

Methodology

Our system is an Online Machine Learning pipeline implemented primarily using the `river` Python library [3], extended with custom components for vectorization and unsupervised drift detection. The pipeline processes one article at a time in strict temporal order, with a test-then-train evaluation approach.

Data Acquisition and Preprocessing

GDELT-based real-world stream. We query the GDELT API v2 [4], [5] using political keywords filtered to left-leaning outlets (e.g., `thenation.com`, $y = 0$) and right-leaning outlets (e.g., `breitbart.com`, $y = 1$) from January 2024 until April 2026. Full article text is extracted using `trafilatura` [6].

Balancing the stream. The acquired dataset is highly imbalanced due to a problem with querying left-wing websites. It consists of $2,231$ articles from left-leaning websites and $12,508$ from right-leaning ones. To produce a more balanced dataset for our experiments, we propose three balancing strategies:

- **Extract max n articles per class per day**: may produce an unbalanced stream in case of days without articles from one of the classes; incurs a small loss of data.
- **Under-sample the majority class**: produces a balanced stream but with a significant loss of data.
- **Split articles in minority class**: keeps all articles from the majority side and splits minority-class articles into parts so that the number of parts matches the majority count; produces a balanced stream without any data loss.

Synthetic data stream. We developed a custom class with two politically-coded vocabularies (V_L, V_R), each containing 50 terms plus a shared 130-word neutral vocabulary. Each synthetic article is generated as a bag of words: 50% from the true-class vocabulary, 10% cross-contamination, and 40% neutral noise. Concept drift is simulated by swapping words between vocabularies. The generator supports three configurable drift types: abrupt (instantaneous swap at step t), gradual (distributed swap between t_{start} and t_{end}), and recurring (vocabulary restoration).

Dynamic Text Vectorization

Online classification of text requires feature extraction that can handle an evolving and unbounded vocabulary without access to the full corpus upfront. We evaluate two fundamentally different approaches.

Incremental TF-IDF. The `river` library's `TFIDF` component maintains running term statistics, producing sparse vectors. TF-IDF is a lightweight bag-of-words baseline blind to word order and semantic similarity.

Dense LLM embeddings. To capture deep contextual relationships, we integrate pre-trained Large Language Models to map raw text into continuous, dense vector spaces. We evaluate two models: the general-purpose `all-MiniLM-L6-v2` Sentence Transformer [7], which produces a 384-dimensional representation, and `bucketresearch/politicalBiasBERT` [8], a domain-adapted model explicitly fine-tuned on political discourse, which yields a 768-dimensional embedding. To process the continuous data stream, we created a custom bridge class inheriting from `river.base.Transformer`. This module encodes incoming articles into named features on the fly, enabling the online classifier to reason about semantic similarities and ideological nuances even as the underlying vocabulary evolves.

Part I: Adaptive Streaming Classifiers

In this phase of the study, we evaluate two contrasting online classification pipelines designed to process a continuous stream of news articles. The primary objective of both models is binary sequence classification: predicting whether an incoming article originates from a “Left” or “Right” leaning news source based solely on its text.

TF-IDF pipeline. The baseline vectorizer is composed with a `MultinomialNB` classifier, the standard choice for sparse count-based features. This pipeline was evaluated on both the synthetic stream and the real-world GDELT corpus. Drift detection used the ADWIN (Adaptive Windowing) algorithm, monitoring per-sample binary classification error.

LLM pipeline. The `StreamSentenceTransformer` is composed with `StandardScaler` (normalizing dimensions online) and `LogisticRegression`. Unlike tree-based methods with grace periods, SGD updates immediately from the first observation, essential in low-volume streams.

Majority-class baseline. For each position in the stream, we identify the most frequent label within the preceding 200 observations and compute the accuracy of predicting that class, distinguishing genuine learning from degenerate behavior.

ADWIN Drift Detector. To continuously monitor the performance of our streaming classifiers and detect concept drift, we employ the ADWIN algorithm [9] for the sequence of binary classification errors (0 for a correct prediction, 1 for an incorrect one).

Drift Detection and Shadow Model Evaluation. For the synthetic experiments, we implement a *reset-on-drift* strategy using the ADWIN algorithm: upon an alert, the active model is re-instantiated fresh, while the previous model is retained as a *shadow model* evaluated in parallel on the incoming stream. To analyze the nature of the drift and mitigate real-world scenarios, we employ two distinct shadow model updating strategies:

- **Frozen Shadow Models:** The model's weights are locked at the exact moment of the drift alert. This approach imitates the situation of cycles in politics (e.g., Trump's second presidency after a break. A model trained only on data from his first presidency could become relevant again during his second term).

- **Adaptive Shadow Models:** The old model continues to train incrementally alongside the newly initialized active model.

Part II: Unsupervised Distributional Drift Detection

The second part takes a complementary approach by abandoning the requirement for accurate binary classification. Instead, we detect drift by measuring semantic shifts in aggregated news streams without relying on class labels.

Time-windowed stream aggregation. Instead of processing articles as individual instances, we aggregate them into daily and weekly time windows. For each window (daily, weekly) and stream group (mixed, left-leaning, and right-leaning), article texts are concatenated into a single document representing the media discourse in that period. This aggregation reduces article-level noise and enables the analysis of broader temporal shifts in political news narratives. For the mixed stream, we additionally evaluate four sampling schemes to reduce the effect of class imbalance: the raw, unbalanced stream, max-n articles per class per day, global undersampling, and minority-class splitting per day.

TF-IDF and embedding-based distance computation. Each aggregated window document is represented in two feature spaces: a lexical TF-IDF space and a semantic embedding space. In the TF-IDF variant, all window documents are vectorized using term frequency–inverse document frequency weights, which capture changes in word usage across time. In the embedding-based variant, each window document is encoded with the `all-MiniLM-L6-v2` Sentence Transformer, producing a 384-dimensional semantic representation.

For both representations, cosine distance is computed between consecutive window vectors, yielding a univariate *drift score stream*:

$$d_t = 1 - \cos(\mathbf{x}_t, \mathbf{x}_{t+1}),$$

where $\mathbf{x}_t$ denotes either the TF-IDF vector or the embedding vector for window t. High cosine distances indicate substantial shifts in the lexical or semantic composition of political discourse.

Multiple drift detectors Three drift detectors from the `river` library are applied independently to the cosine distance stream:

- **ADWIN (Baseline)**: Detects distributional shifts using an adaptive sliding window with $\delta = 0.01$ and clock $= 1$.
- **Page-Hinkley**: Monitors cumulative deviations from the baseline mean, with $\delta = 0.005$ and $\alpha = 0.999$. For daily streams, we use threshold $= 0.15$ and min_instances $= 5$; for weekly streams, we use threshold $= 0.10$ and min_instances $= 3$.
- **KSWIN**: Uses a Kolmogorov–Smirnov windowing test with $\alpha = 0.01$. For daily streams, we set window_size $= 20$ and stat_size $= 5$; for weekly streams, we set window_size $= 8$ and stat_size $= 3$.

Drift detections are matched against known political events (election, inauguration, debates, trials, etc.) within a tolerance window of 7–14 days, depending on whether daily or weekly aggregation is used, respectively.

Conducted Experiments and Results

Synthetic Streams Classification

The first part of the experiment checks whether a classification-based approach to concept drift works with natural language data. We first evaluate our approach on synthetically generated streams with known drift patterns. This controlled setting allows us to verify that our drift detectors correctly identify injected concept drifts and assess their performance across different drift types.

Basic drift simulation. A stream of 3,200 articles is generated with abrupt drifts at steps 800, 1,600, and 2,400 (swapping 50% of vocabulary). The TF-IDF + MultinomialNB pipeline with a reset-on-drift strategy (all models continue training) shows rolling accuracy climbing during stable segments, dropping sharply at each drift point, and recovering after ADWIN detection and model reset (Figure 1).

Advanced drift simulation. A 4,000-article stream features: (1) abrupt drift at step 800 (swap 50%), (2) gradual drift spanning steps 1,500–2,000 (swap 90%), and (3) recurring concept at step 3,000 (restore original). This time we froze shadow models at replacement to simulate cycles in politics. After restoring the original vocabulary (at step 3,000), the frozen model achieves perfect accuracy faster than a newly trained one. However, gradual drift does not trigger any alerts, possibly due to slow changes in vocabulary (Figure 2).

Real-World Stream Classification

Next, we validate our methods on real-world news articles from the GDELT database, spanning the

2024 U.S. Presidential Election through early 2026. Since the raw GDELT dataset is severely imbalanced (2,231 left-leaning vs. 12,508 right-leaning articles), we systematically address this challenge by applying the **Text Splitting for Minority Class** strategy described in the Methodology section. This approach divides lengthy minority-class articles into independent chronological segments without data loss, generating a perfectly balanced stream of 20,446 text segments (10,233 per class). This larger corpus not only mitigates class imbalance but also provides the volume necessary for robust online learning in a classification task. The dataset spans critical periods including the campaign phase, the general election, and the initial months of governance, including significant semantic shifts in political discourse.

TF-IDF Word Representation. Evaluating the TF-IDF + MultinomialNB pipeline on the balanced stream demonstrates strong predictive capabilities. As shown in Figure 3, the rolling majority-class baseline (red solid line) stabilizes at 0.50, while the streaming classifier (blue solid line) maintains a rolling accuracy between 85–95% throughout the evaluation period. This substantial margin above the baseline confirms that sparse, keyword-based TF-IDF features possess significant discriminative power for distinguishing polarized political rhetoric.

Critically, the ADWIN drift detector successfully monitored the error stream and triggered multiple alerts (red dashed lines) that correspond to major political events. These detections demonstrate that the detector can identify genuine semantic shifts in the news stream. Notably, ADWIN detected the start of the campaign and intensified around the governance transition, with particularly strong signals after the inauguration and perfectly at the 100-days-in-office milestone, suggesting that vocabulary-level changes are most pronounced during major institutional shifts.

Dense LLM Embeddings Representation. To evaluate the efficacy of deep semantic representations, we applied the dense embedding pipeline (`StreamSentenceTransformer` + SGD Logistic Regression) to the same balanced stream. This approach also demonstrates robust predictive power, maintaining a rolling accuracy between 80–90% well above the 0.50 baseline (Figure 4).

A major difference lies in concept drift detection, where TF-IDF proves to be significantly more effective. While it successfully captured multiple narrative shifts during the government transition, the LLM pipeline detected only one (mid-2024). This happens because LLMs process text by broad meaning, making them highly resistant to vocabulary changes. While this stability is excellent for general classification, it makes the LLM too rigid to catch fast-paced changes in political jargon. Ultimately, if the goal is to actively detect concept drift and track the evolving daily news cycle, TF-IDF is the superior tool, whereas LLMs are better suited for maintaining a stable, long-term classifier.

Changing the embedding model for `bucketresearch/politicalBiasBERT` showed even higher performance (above 90%) but without noticable change in drift detection capabilities (Figure 5).

Accuracy comparison. A combined analysis of the models' rolling accuracy (Figure 6) over the normalized timeline reveals that the performance is heavily influenced by the chosen preprocessing strategies. Configurations that applied text splitting to balance the minority class achieved the highest sustained performance. Conversely, heavily restricting the training data to a maximum of three samples per class per day significantly degraded performance.

Unsupervised Distributional Drift Detection

The second part of the experiment evaluates whether sudden political events can be detected directly from changes in the textual distribution of news articles, without relying on supervised classification accuracy. Instead of classifying individual articles, the method compares consecutive daily or weekly aggregated text windows using cosine distance in TF-IDF and embedding-based representation spaces. The resulting distance streams are then monitored with ADWIN, Page-Hinkley, and KSWIN drift detectors.

Overall detector behavior. The results show that *Page-Hinkley* is the most effective detector for this task. It consistently identifies abrupt upward changes in the cosine-distance streams, which corresponds well to sudden changes in political discourse. This behavior is especially visible in the daily TF-IDF mixed stream shown in Figure 7, where multiple detector alarms occur close to annotated political events. In contrast, ADWIN does not detect meaningful drift under the selected configuration, making it too conservative for this setting. KSWIN detects only a small number of changes and therefore provides weaker event coverage. Consequently, the main interpretation focuses on Page-Hinkley, while ADWIN and KSWIN are treated as baseline or supporting detectors.

Best-performing configurations. The strongest overall event coverage is obtained for the daily raw mixed stream represented with TF-IDF and monitored with Page-Hinkley. As shown in Figure 7, this configuration is highly sensitive to abrupt lexical changes and detects all annotated political events, reaching an event recall of 1.00. However, this sensitivity also leads to many false positives: 33 detections in total, including 14 true positives and 19 false positives, resulting in a precision-like score of 0.424. Therefore, this setup is useful when maximizing event coverage is more important than minimizing false alarms.

A more balanced result is obtained using weekly aggregation with embedding-based representations on the mixed stream sampled with max-3 articles per class per day. Figure 8 shows that this configuration produces a much cleaner drift signal, detecting 8 out of 10 events while producing only one false positive. Its precision-like score of 0.889 is the highest among the high-recall configurations. This suggests that weekly aggregation reduces short-term noise, while class-balanced semantic representations preserve meaningful event-related changes.

TF-IDF versus embedding-based representations. TF-IDF performs best in the daily raw mixed stream, where it captures rapid lexical changes associated with the political calendar. The dense cluster of alarms in Figure 7 confirms that lexical shifts are strongly reflected in consecutive daily TF-IDF vectors. Its main advantage is sensitivity: all major annotated events are detected within the accepted tolerance window. However, this comes at the cost of a relatively high number of false alarms.

Embedding-based representations are less effective in the raw daily setting, but become highly competitive after balancing the mixed stream. The daily balanced embedding configuration in Figure 9 reaches an event recall of 0.90, showing that semantic representations can still capture event-related discourse changes when class imbalance is reduced. Compared with the weekly embedding setup in Figure 8, the daily variant provides better temporal resolution but produces more false positives.

Daily versus weekly aggregation. Daily aggregation provides better temporal resolution and captures short-lived discourse changes around individual events. This is visible in Figure 7, where alarms appear around almost all major campaign events. However, daily aggregation also increases sensitivity to temporary fluctuations in the news cycle.

Weekly aggregation smooths the drift score stream and reduces short-term volatility. Although this lowers event recall compared with the best daily setup, it substantially improves precision-like behavior. This trade-off is illustrated by Figure 8, where the weekly embedding-based stream produces fewer alarms while maintaining strong alignment with major events. Therefore, the weekly embedding-based mixed max-3 configuration is the most reliable compromise between interpretability and event coverage.

Effect of stream composition and sampling. The mixed stream generally provides the strongest event coverage because it captures changes across the whole political media environment. However, the raw mixed stream is imbalanced, with substantially more right-leaning than left-leaning articles. This imbalance makes the raw mixed stream highly reactive, as seen in Figure 7, but also increases the number of false positives.

To reduce this effect, three additional mixed-stream sampling schemes are evaluated: max-n articles per class per day, global undersampling, and minority-class splitting per day. The max-3 per class per day strategy gives the best balanced results, especially for embedding-based representations. This can be seen by comparing the daily balanced embedding stream in Figure 9 with the cleaner weekly version in Figure 8. Both plots show that class balancing helps embeddings capture meaningful semantic changes without being dominated by the majority class.

Ideology-specific drift patterns. The ideology-specific streams reveal an asymmetric pattern. The right-leaning stream produces much stronger event coverage than the left-leaning stream. With TF-IDF and daily aggregation, the right-only stream detects 8 out of 10 events, as shown in Figure 10. The alarms are especially visible around events directly related to Donald Trump and the Republican campaign. This suggests that ideology-specific streams can reveal which side of the media ecosystem reacts more strongly to particular political events.

Interpretation. Overall, the results confirm that unsupervised distance-based drift detection can capture meaningful changes in political news discourse. The best configuration for maximum event coverage is daily TF-IDF on the raw mixed stream, shown in Figure 7. In contrast, the best configuration for a cleaner and more reliable signal is weekly embedding-based detection on the class-balanced mixed stream, shown in Figure 8. These findings suggest that lexical representations are more sensitive to abrupt event-driven vocabulary shifts, whereas embedding-based representations combined with weekly aggregation provide a more stable view of semantic change.

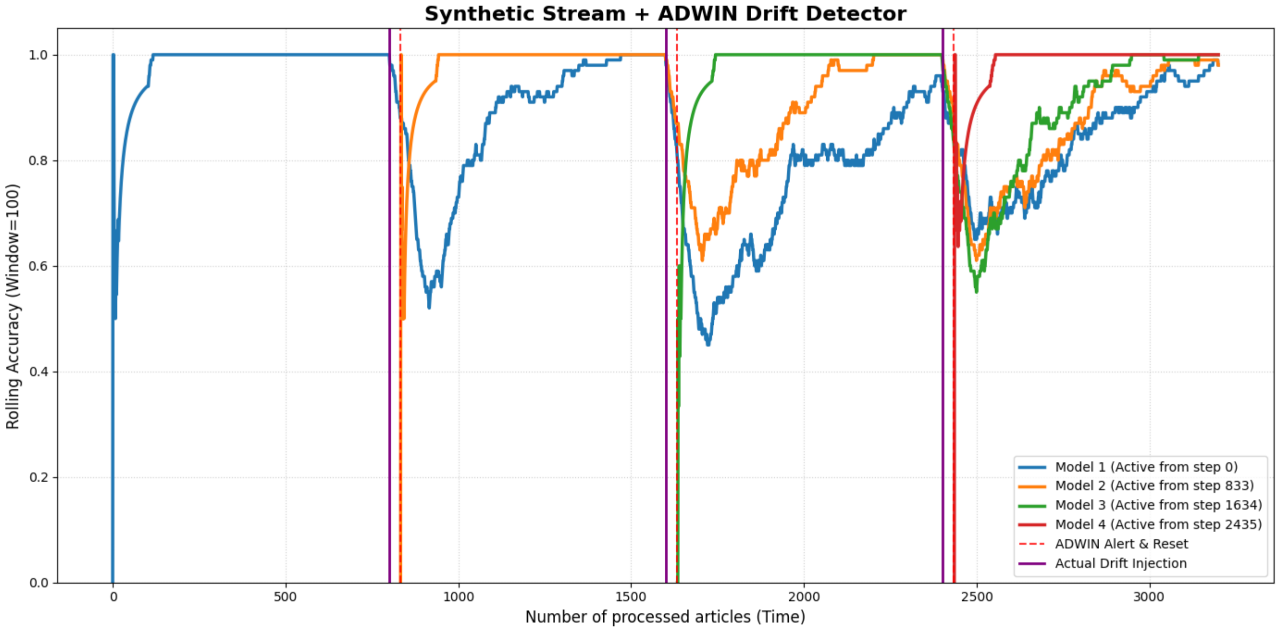

Figure 1: *Performance on the basic synthetic stream with three abrupt concept drifts. ADWIN alerts (red dashed lines) closely follow each injected drift (purple solid lines) and trigger model resets. Fresh models (later-colored traces) recover to near-100% accuracy within hundreds of steps, while frozen shadow models (earlier traces) show degraded performance after each subsequent drift event.*

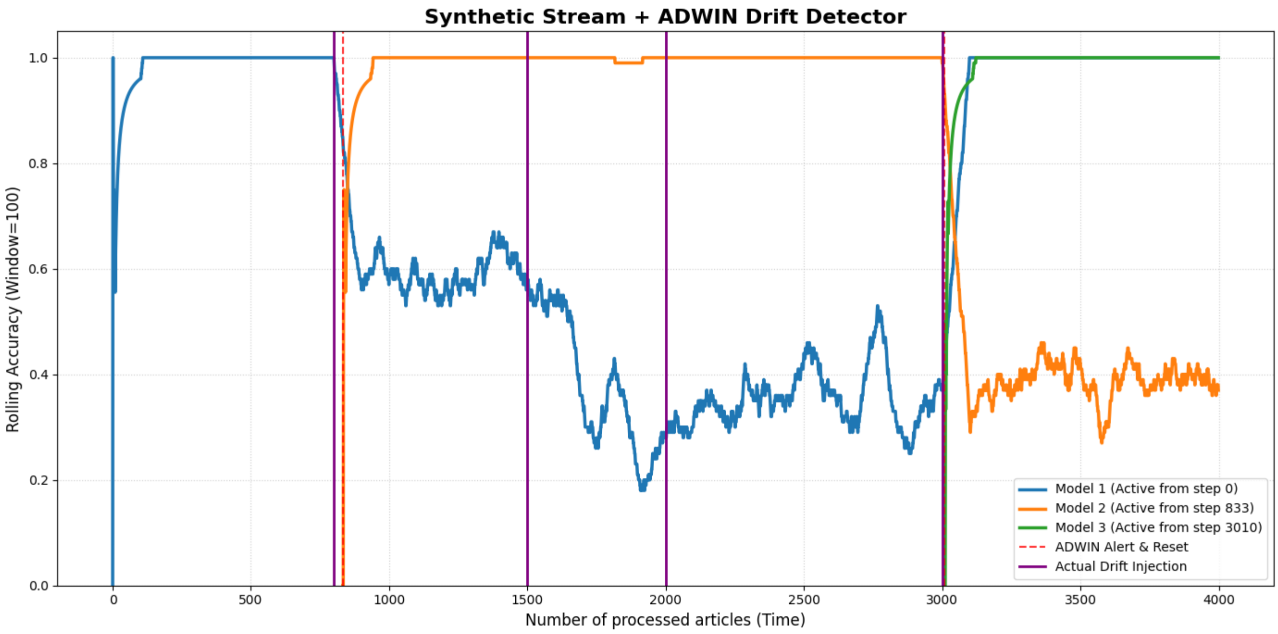

Figure 2: *Performance on the advanced synthetic stream with mixed drift types. The model encounters an abrupt drift at step 800, a gradual drift spanning steps 1,500–2,000, and a recurring concept drift at step 3,000 (vocabulary restoration). ADWIN detection is less responsive to gradual drift due to slow evidence accumulation, but sharply alerts on the recurring drift when pre-adapted weights fail against restored original vocabulary.*

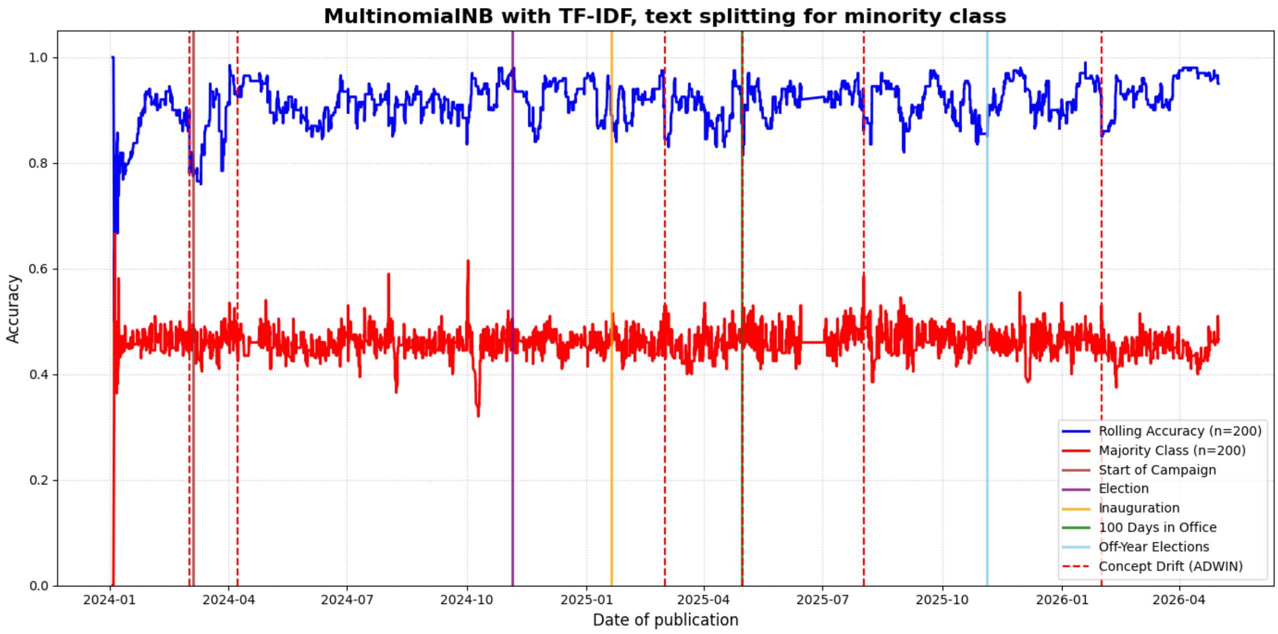

Figure 3: *Performance and concept drift detection of the TF-IDF + MultinomialNB pipeline on the balanced real-world stream (20,446 articles). The model's rolling accuracy (blue line, $n = 200$) consistently outperforms the 0.50 majority-class baseline (red line), proving the presence of a strong discriminative signal. ADWIN drift alerts (red dashed vertical lines) demonstrate semantic shifts in political discourse, notably detecting Start of Campaign and avoiding Election Day but triggering heavily during the transition to active governance (shortly after Inauguration and precisely at 100 Days in Office).*

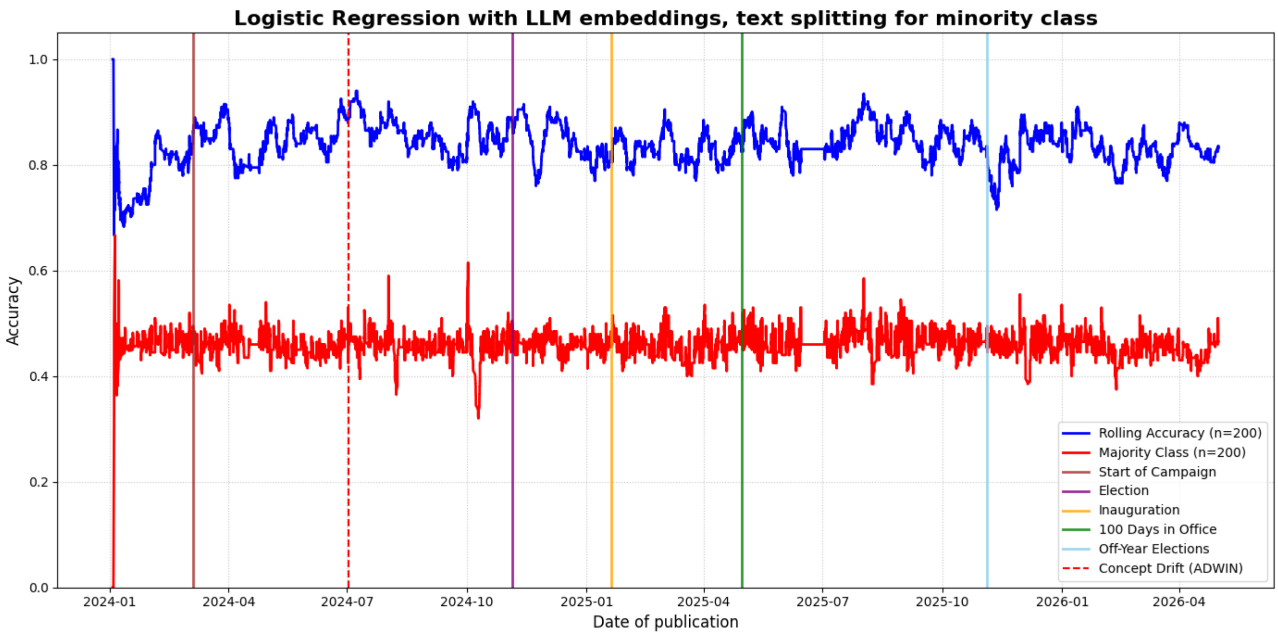

Figure 4: *Performance and concept drift detection of the LLM Embeddings + Logistic Regression pipeline on the balanced real-world stream (20,446 articles). The model successfully learns the decision boundary, maintaining a rolling accuracy (blue line, $n = 200$) of 80–90% well above the trivial 0.50 baseline (red line). Notably, ADWIN detects only a single concept drift (red dashed line in mid-2024).*

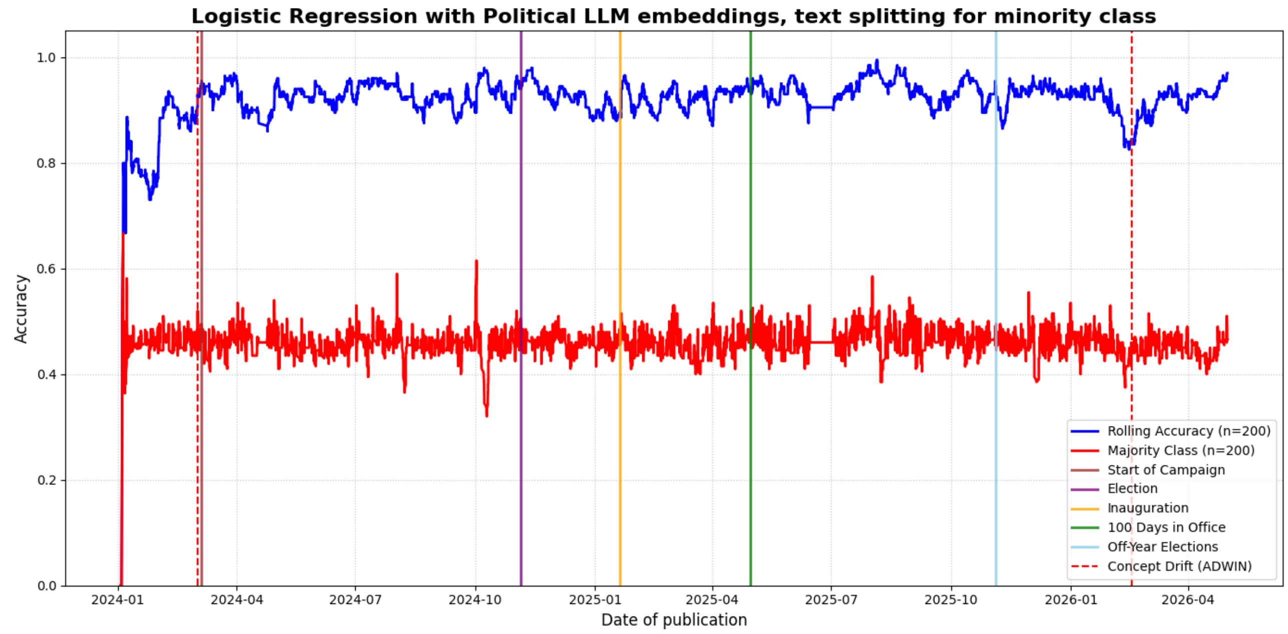

Figure 5: *Performance and concept drift detection of the Political LLM Embeddings + Logistic Regression pipeline on the balanced real-world stream (20,446 articles). The model achieved higher rolling accuracy (blue line, $n = 200$) than the base embedding model (around 90% compared to around 80%) while detecting one more drift (two in total, one of them is aligned with the start of the campaign).*

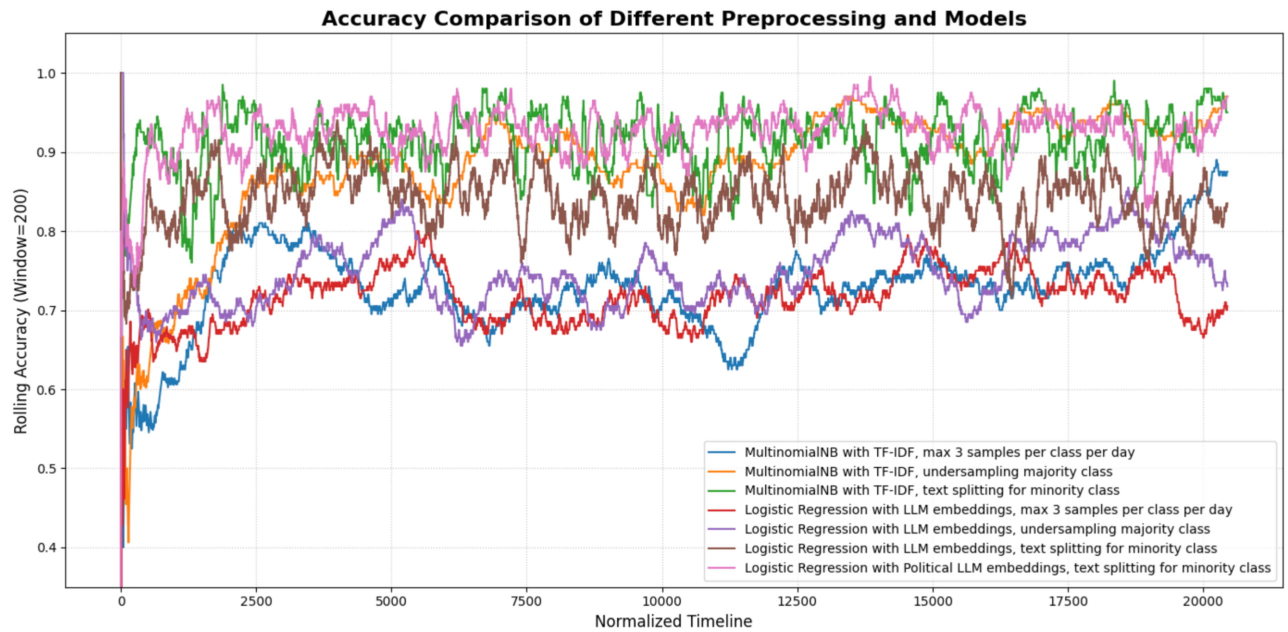

Figure 6: *Rolling accuracy (using a window size of 200) of seven distinct machine learning models and preprocessing configurations over a normalized timeline. Models employing text splitting to balance the minority class consistently demonstrate the highest accuracy, largely fluctuating between 0.85 and 0.98. In contrast, configurations strictly limiting the data to a maximum of three samples per class per day result in the lowest overall performance, generally hovering between 0.65 and 0.80. The highest rolling accuracy was achieved by MultinomialNB with TF-IDF and Logistic Regression with Political LLM, both with text splitting for minority class strategy.*

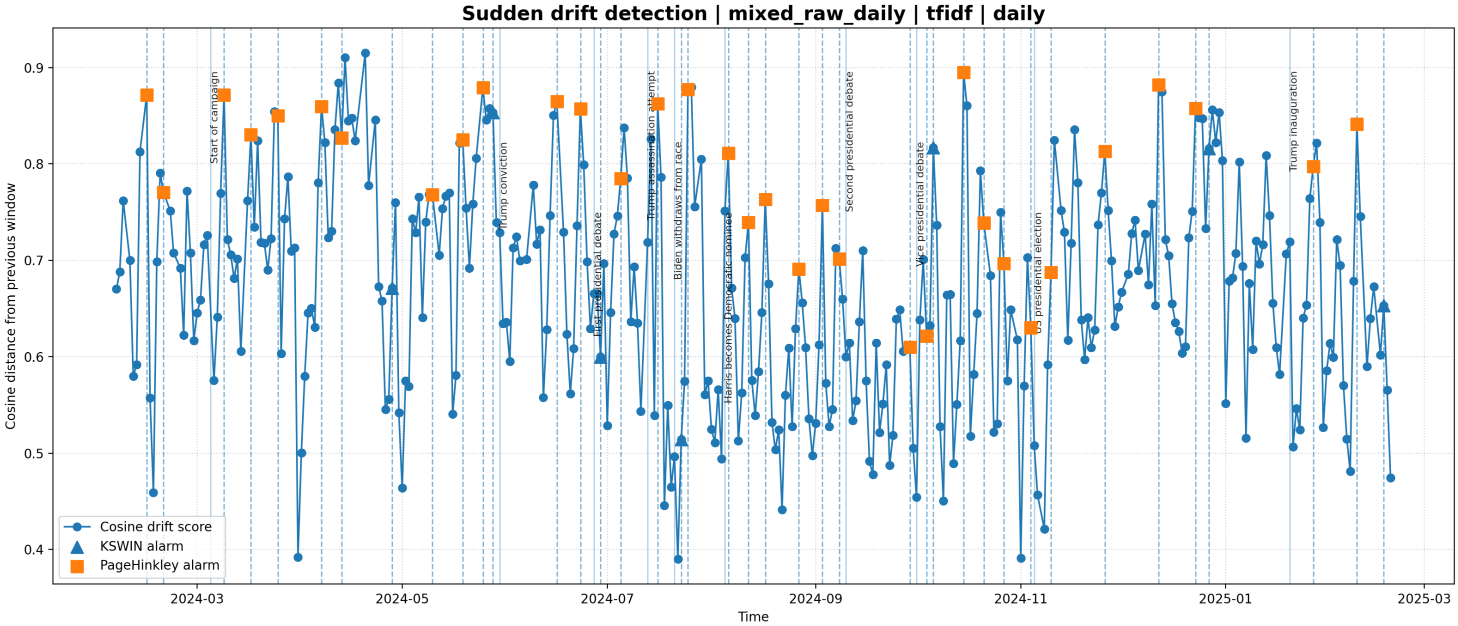

Figure 7: *Daily TF-IDF drift detection on the raw mixed stream. This configuration achieves the highest event recall, detecting all annotated political events, but also produces several false positives due to the high sensitivity of daily lexical changes.*

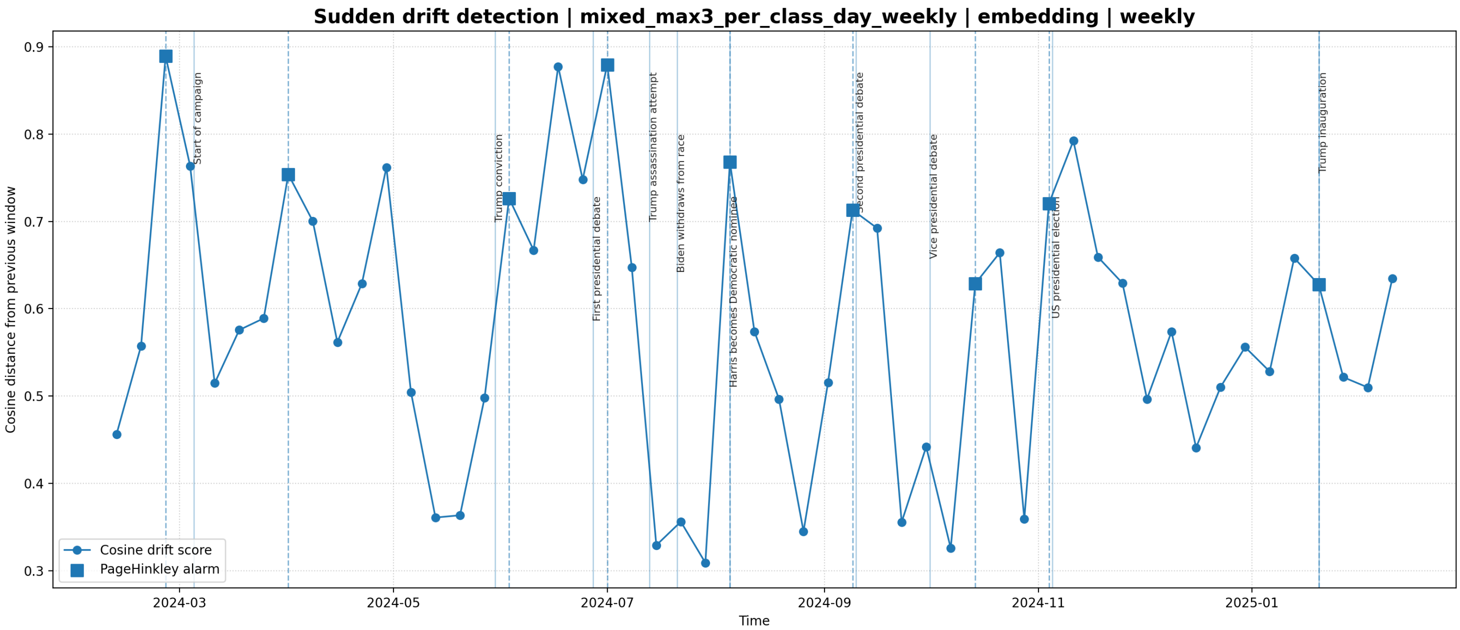

Figure 8: *Weekly embedding-based drift detection on the mixed stream with max 3 articles per class per day. This configuration provides the best precision-like trade-off, detecting 8 out of 10 events with only one false positive.*

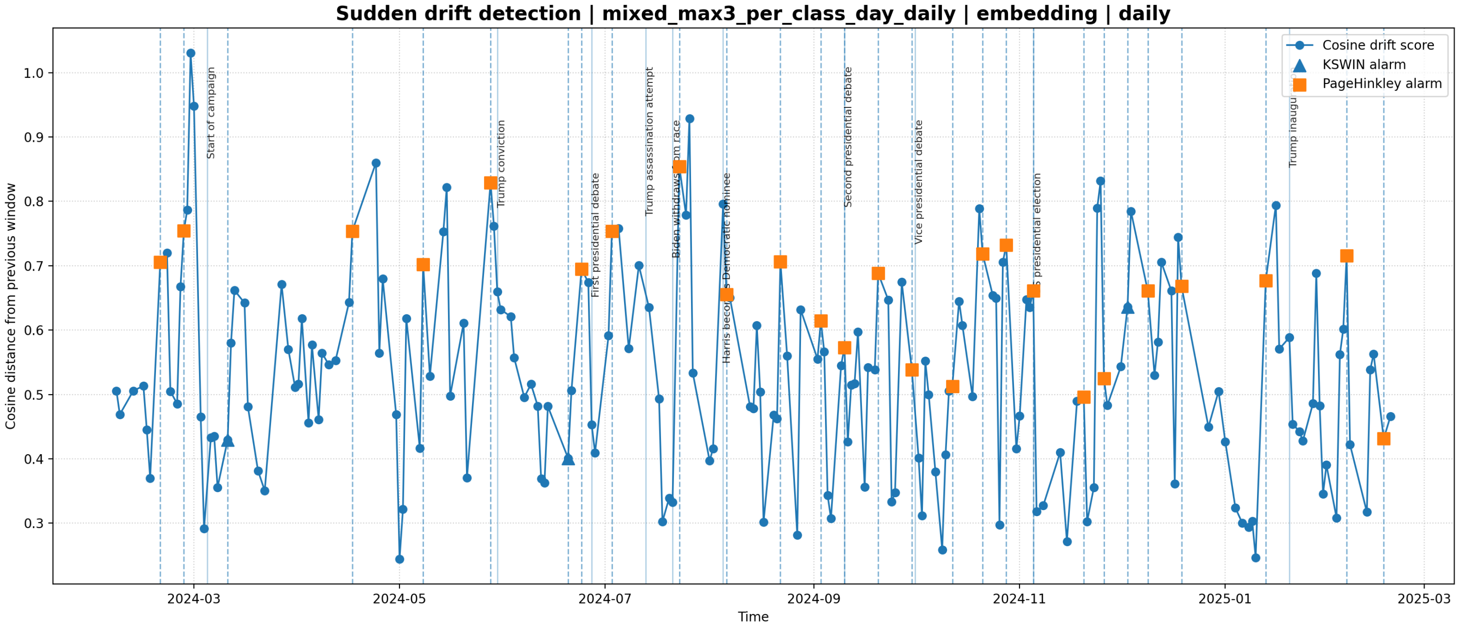

Figure 9: *Daily embedding-based drift detection on the mixed stream balanced with max-3 articles per class per day. The result shows strong event coverage while reducing the influence of class imbalance compared with the raw mixed stream.*

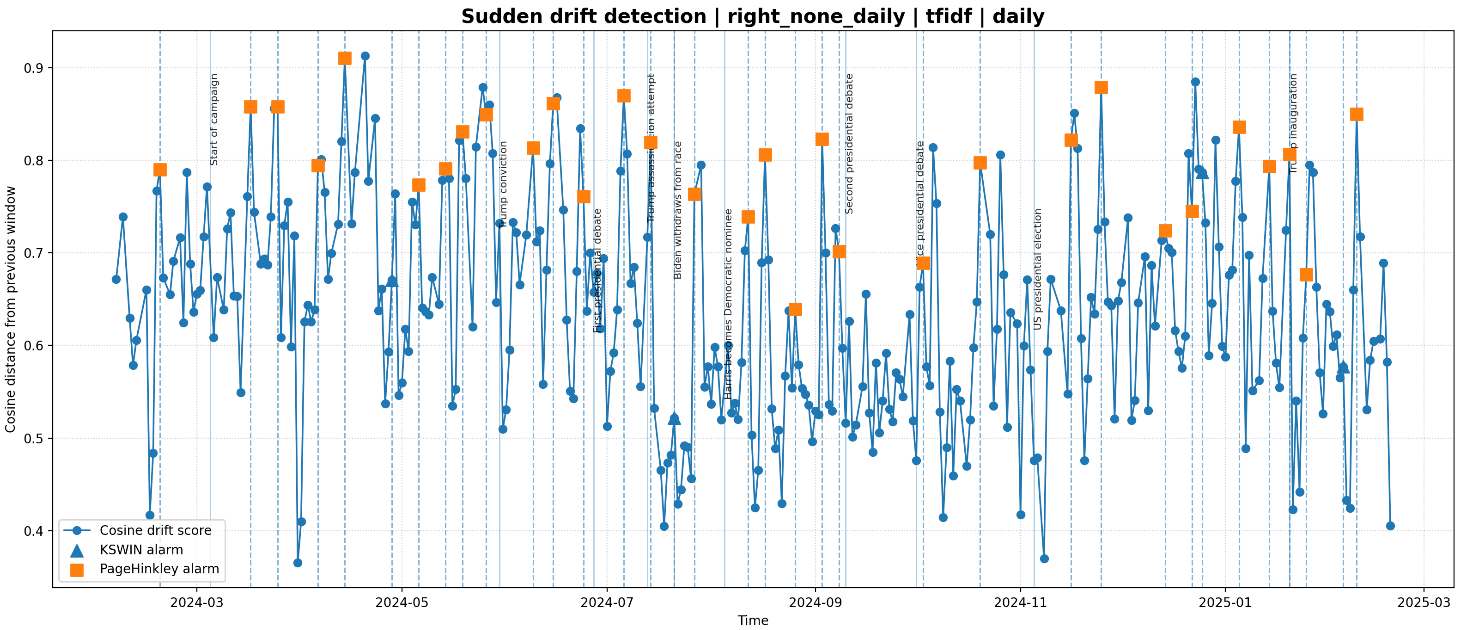

Figure 10: *Daily TF-IDF drift detection on the right-leaning stream. The stream shows strong sensitivity to major campaign events, especially those related to Donald Trump and the Republican campaign, illustrating ideology-specific drift behavior.*

Discussion

Key Findings

Part I: Adaptive Streaming Classifiers. The experiments demonstrate that both sparse and dense streaming classifiers can successfully categorize polarized political text, provided that extreme class imbalance is mitigated. By implementing a text-splitting strategy to balance the stream (expanding it to over 20,000 instances), both models (TF-IDF + MultinomialNB and LLM Embeddings + Logistic Regression) successfully detached from the 0.50 majority-class baseline. This proves that both keyword-based and deep semantic feature spaces contain a strong discriminative signal for political discourse.

The most significant finding lies in the contrasting drift detection behaviors of the two architectures. While our synthetic experiments established that ADWIN reliably detects structural data shifts, applying it to the real-world stream revealed differences between lexical sensitivity and semantic stability. The sparse vectorizer (TF-IDF) proved to be more sensitive, which caused more drift detections than the more stable LLM embeddings.

Part II: Unsupervised Distributional Drift Detection. In the unsupervised setting, drift is detected directly from changes in the textual distribution. Daily TF-IDF on the raw mixed stream gives the highest event sensitivity, detecting all annotated political events, but also produces many false positives. In contrast, weekly embedding-based detection with max-n articles per class per day provides a cleaner and more stable signal, detecting most events with far fewer false alarms.

The ideology-specific analysis further shows that left- and right-leaning streams react differently to political events. Right-leaning streams show stronger drift around Trump- and Republican-related events, while left-leaning streams better reflect Democratic-specific moments such as Biden's withdrawal and Harris's nomination.

Conclusions

Supervised drift detection is useful for monitoring model degradation, while unsupervised distributional drift detection provides more interpretable evidence of changes in political discourse. The best overall strategy is therefore to combine both views: use supervised methods to track predictive stability and unsupervised distance-based methods to identify event-driven changes in the media narrative.

Limitations

Our experiments showed that the larger the data corpus is, the better and more stable the results are. Therefore, it would be beneficial to gather even more articles, potentially using tools other than GDELT, which we believe we have extracted the maximum available data from. Moreover, the binary left/right categorization could be oversimplified; real media bias varies by outlet, author, and topic.

Future Work

Based on the current findings, future research could focus on two primary directions:

- **Hybrid Streaming Ensembles:** Developing an architecture that integrates the strengths of both vectorization methods. Such a system would leverage the semantic stability of domain-adapted LLMs for the core classification task, while operating a parallel TF-IDF module exclusively as a highly sensitive novelty detector to capture rapid narrative shifts.
- **Explainable Concept Drift (XAI):** Transitioning from only detecting drift to explaining it. Integrating online Explainable AI techniques into the semantic pipeline would allow researchers to automatically extract and interpret the specific ideological narratives or vocabulary changes that triggered the anomaly.

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